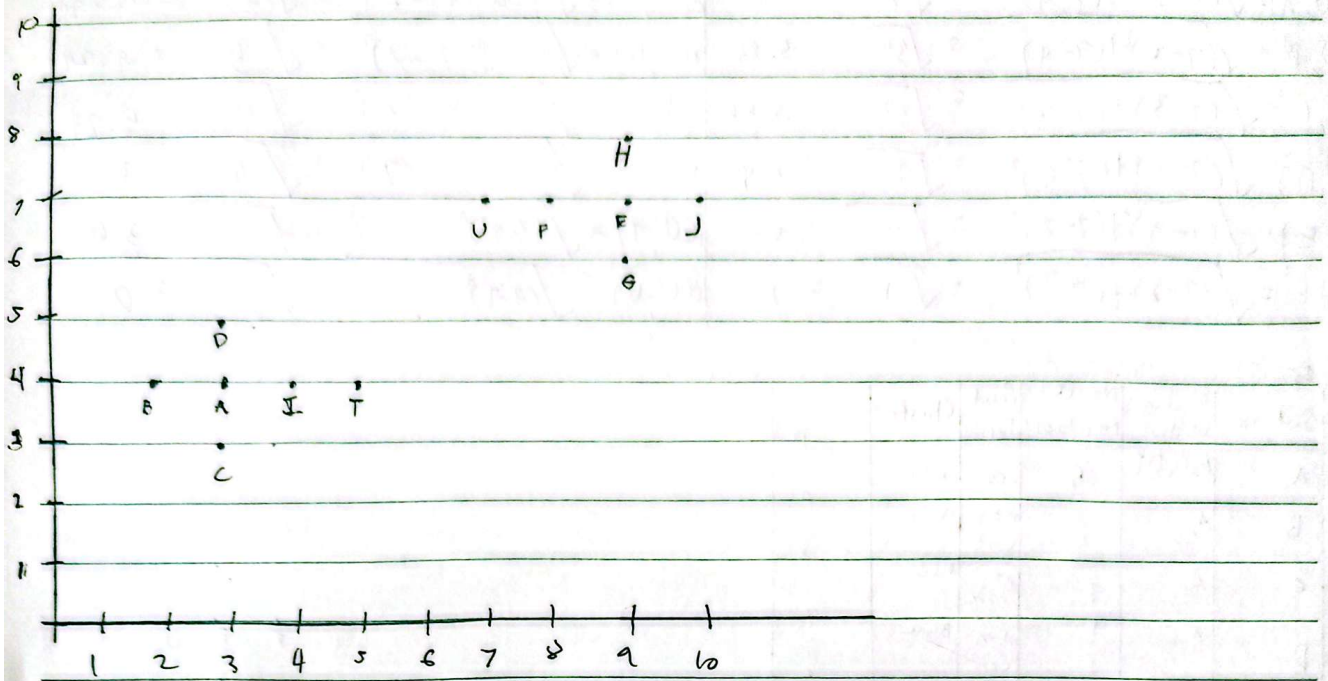




	(3,4) A	(2,4) B	(3,3) C	(3,5) D	(5,7) E	(6,7) F	(7,7) G	(8,7) H	(9,7) I	(10,7) J	(5,4) K	(6,4) L	(7,4) M	(8,4) N	(9,4) O	(10,4) P	(5,3) Q	(6,3) R	(7,3) S	(8,3) T	(9,3) U	(10,3) V
A	0	1	1	1	6.7	5.8	6.3	7.2	1	7.6	2	5										
B	1	0	1.4	1.4	7.6	6.7	7.2	2	2	8.5	2	5.83										
C	1	1.4	0	2	7.2	6.4	6.7	7.2	1.4	8	2.24	5.66										
D	1	1.4	2	0	6.3	5.3	6	6.7	1.4	7.2	2.24	4.47										
E	6.7	7.6	7.2	6.3	0	1	1	1	5.8	1	5	2										
F	5.8	6.7	6.4	5.3	1	0	1.4	1.4	5	2	4.47	1										
G	6.3	7.2	6.7	6	1	1.4	0	2	5.3	1.4	4.47	2.24										
H	7.2	2	7.2	6.7	1	1.4	2	0	6.4	1.4	5.66	2.24										
I	1	2	1.4	1.4	5.8	5	5.3	6.7	0	6.7	1	4.24										
J	7.6	8.5	8	7.2	1	2	1.4	1.4	6.7	0	5.83	3										
K	2	2	2.24	2.24	5	4.47	4.47	5.66	1	5.83	0	3.61										
L	5	5.83	5.66	4.47	2	1	2.24	2.24	4.24	3	3.61	0										

$$\begin{aligned}
 d(T, A) &= \sqrt{(5-3)^2 + (4-4)^2} = \sqrt{4} = 2 \\
 d(T, B) &= \sqrt{(5-2)^2 + (4-4)^2} = \sqrt{9} = 3 \\
 d(T, C) &= \sqrt{(5-3)^2 + (4-3)^2} = \sqrt{5} = 2.24 \\
 d(T, D) &= \sqrt{(5-3)^2 + (4-5)^2} = \sqrt{5} = 2.24 \\
 d(T, E) &= \sqrt{(5-1)^2 + (4-7)^2} = \sqrt{25} = 5 \\
 d(T, F) &= \sqrt{(5-7)^2 + (4-8)^2} = \sqrt{20} = 4.47 \\
 d(T, G) &= \sqrt{(5-7)^2 + (4-8)^2} = \sqrt{20} = 4.47 \\
 d(T, H) &= \sqrt{(5-8)^2 + (4-8)^2} = \sqrt{32} = 5.66 \\
 d(T, I) &= \sqrt{(5-9)^2 + (4-7)^2} = \sqrt{34} = 5.83 \\
 d(T, J) &= \sqrt{(5-10)^2 + (4-7)^2} = \sqrt{34} = 5.83 \\
 d(T, K) &= \sqrt{(5-5)^2 + (4-4)^2} = \sqrt{0} = 0 \\
 d(T, L) &= \sqrt{(5-6)^2 + (4-7)^2} = \sqrt{10} = 3.16 \\
 d(T, M) &= \sqrt{(5-7)^2 + (4-7)^2} = \sqrt{10} = 3.16 \\
 d(T, N) &= \sqrt{(5-8)^2 + (4-7)^2} = \sqrt{18} = 4.24 \\
 d(T, O) &= \sqrt{(5-9)^2 + (4-7)^2} = \sqrt{34} = 5.83 \\
 d(T, P) &= \sqrt{(5-10)^2 + (4-7)^2} = \sqrt{34} = 5.83 \\
 d(T, Q) &= \sqrt{(5-5)^2 + (4-3)^2} = \sqrt{1} = 1 \\
 d(T, R) &= \sqrt{(5-6)^2 + (4-3)^2} = \sqrt{2} = 1.41 \\
 d(T, S) &= \sqrt{(5-7)^2 + (4-3)^2} = \sqrt{5} = 2.24 \\
 d(T, T) &= \sqrt{(5-8)^2 + (4-3)^2} = \sqrt{10} = 3.16 \\
 d(T, U) &= \sqrt{(5-9)^2 + (4-3)^2} = \sqrt{18} = 4.24 \\
 d(T, V) &= \sqrt{(5-10)^2 + (4-3)^2} = \sqrt{25} = 5
 \end{aligned}$$

$$\begin{aligned} \delta(U, A) &= \sqrt{(7-3)^2 + (7-4)^2} = \sqrt{25} = 5 \\ \delta(U, B) &= \sqrt{(7-2)^2 + (7-4)^2} = \sqrt{37} = 5.83 \\ \delta(U, C) &= \sqrt{(7-3)^2 + (7-3)^2} = \sqrt{32} = 5.66 \\ \delta(U, D) &= \sqrt{(7-3)^2 + (7-5)^2} = \sqrt{20} = 4.47 \\ \delta(U, E) &= \sqrt{(7-9)^2 + (7-7)^2} = \sqrt{4} = 2 \\ \delta(U, F) &= \sqrt{(7-8)^2 + (7-7)^2} = \sqrt{1} = 1 \\ \delta(U, G) &= \sqrt{(7-9)^2 + (7-6)^2} = \sqrt{5} = 2.24 \\ \delta(U, H) &= \sqrt{(7-9)^2 + (7-2)^2} = \sqrt{58} = 7.62 \\ \delta(U, I) &= \sqrt{(7-4)^2 + (7-4)^2} = \sqrt{16} = 4 \\ \delta(U, J) &= \sqrt{(7-10)^2 + (7-7)^2} = \sqrt{9} = 3 \\ \delta(U, K) &= \sqrt{(7-5)^2 + (7-4)^2} = \sqrt{17} = 4.12 \\ \delta(U, L) &= \sqrt{(7-7)^2 + (7-3)^2} = \sqrt{16} = 4 \\ \delta(U, M) &= \sqrt{(7-7)^2 + (7-7)^2} = \sqrt{0} = 0 \end{aligned}$$

$\epsilon = 1$ min, $\delta = 2$

Customer	Neighbors within	No. of Neighbors	Point Type	Cluster
A	B, C, D, E	4	Core	Blue
B	A	1	Non Core	Blue
C	A	1	Non Core	Blue
D	A	1	Non Core	Blue
E	F, G, H, I	4	Core	Red
F	E, U	2	Core	Red
G	E	1	Non Core	Red
H	E	1	Non Core	Red
I	A, T	2	Core	Blue
J	E, T	2	Core	Red
K	I	1	Non Core	Blue
L	F	1	Non Core	Red
U	F	1	Non Core	Red

